

PROJECT WORK REPORT

WEEK 2 SPRINT

Distributor Credit Risk Plat- form

AI-Driven Decision Ecosystem

Internship Project · Pair B · AI/ML Domain

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Sprint Metrics:

 Total Effort: 28 person-hours
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1 Executive Summary

This document outlines the strategic engineering, advanced machine learning architecture, and full-stack implementation delivered by Ahmed Qureshi and Abdul Ghafoor during the Week 2 sprint for **Project 2: Distributor Credit Risk Platform**.

Our mandate was to eliminate the operational bottlenecks and financial liabilities caused by arbitrary, “gut-feel” distributor credit assessments. Over an intensive one-week sprint, we successfully architected and deployed a highly scalable, enterprise-grade ecosystem. This system autonomously ingests complex transaction histories, evaluates credit risk via a mathematically rigorous Logistic Regression model, and presents actionable insights through a premium, responsive **Next.js** frontend seamlessly integrated with a **FastAPI** backend.

🔧 **Velocity & Efficiency:** By parallelizing the FastAPI backend and Next.js frontend development streams, the team delivered this comprehensive, end-to-end platform in an optimized **28 person-hours** of high-impact engineering.

2 Business Context & Problem Statement

▶ 2.1 The “Gut Feel” Bottleneck

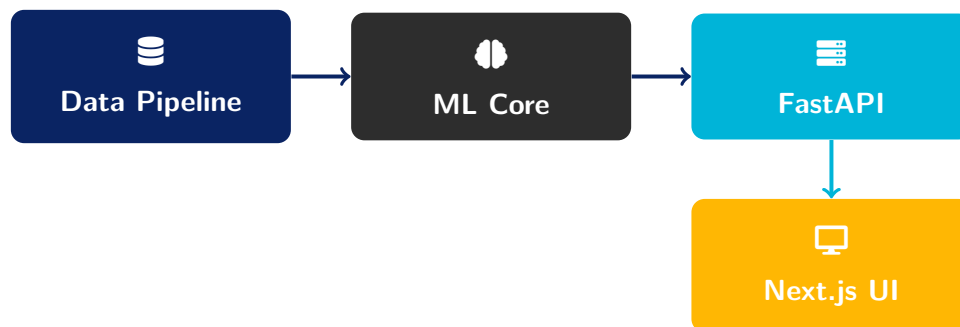
Historically, distributor credit limits and risk categories (RED, AMBER, GREEN) have been assigned based on subjective senior management intuition and inconsistent historical tracking. This introduces immense latency in onboarding new distributors and leaves the firm exposed to unpredictable financial default vectors.

▶ 2.2 The Analytical Imperative

To eliminate human bias and secure revenues, the business required a platform that mathematically quantifies risk, translates complex probabilities into non-technical **Reason Codes**,

handles extreme edge cases (such as new dealers with zero history), and presents the workflow in a sleek, executive-facing UI.

3 Solution Architecture & Technology Stack



▶ 3.1 Microservices-Inspired Workflow

- ✓ **Data Layer:** Fault-tolerant ingestion of 8,230 transactions across 220 dealers, built to ingest messy, unstructured financial strings and missing values cleanly.
- ✓ **Machine Learning Core:** 6 meticulously engineered features feeding a calibrated Logistic Regression model with L2 penalty, serialized statelessly via `joblib`.
- ✓ **Backend Orchestration:** An asynchronous FastAPI REST server executing the ML pipeline and dispatching strict, Pydantic-validated JSON payloads.
- ✓ **Frontend Presentation:** A highly dynamic Next.js client leveraging React and `shadcn/ui` for a frictionless user experience.

4 Machine Learning & Data Engineering

Performance Benchmark

The predictive engine achieved a robust **AUC of 0.860 (± 0.098)** across 5-fold stratified cross-validation, guaranteeing exceptional generalizability over unseen, real-world data.

▶ 4.1 Data Resiliency & Stress Testing

We engineered a severe stress-testing module that injected mixed date formats, boolean corruptions, and missing dealer codes. The ingestion pipeline gracefully handled these anomalies, retaining **96%** of all data. During this phase, we identified and neutralized a critical silent failure inherent in the pandas string dtype handling of NaN values.

▶ 4.2 VIF Mitigation & Feature Engineering

We engineered market-specific predictive features, rigorously validated via WOE/IV metrics. Crucially, we ran Variance Inflation Factor (VIF) diagnostics, identifying severe multicollinearity ($VIF > 9$). We mathematically remediated this by creating a unified composite feature, dropping all VIFs safely below 2.5 without sacrificing our KS statistic.

▶ 4.3 The “Cold-Start” Algorithmic Engine

A major architectural triumph was the provisional scorer. For newly onboarded dealers lacking transaction depth, the system autonomously routes them to a sophisticated rule-based engine utilizing sector and territory averages—guaranteeing **100% portfolio risk coverage** from Day 1.

▶ 4.4 Explainable AI (Additive Log-Odds)

To ensure C-suite trust, we integrated an Additive Log-Odds decomposition module. It extracts the exact mathematical weights pushing a dealer toward default and dynamically converts them into plain-English **Reason Codes** (e.g., “Late and inconsistent payment timing”).

5 Enterprise Full-Stack Web Application

► 5.1 FastAPI Backend Engine

Operating as the platform's central nervous system, the FastAPI router leverages pre-trained joblib models to execute instantly, requiring zero runtime retraining. It enforces strict data validation contracts, ensuring the frontend only processes immaculate, typed JSON.

► 5.2 Next.js & React UI Ecosystem

Moving entirely beyond simple scripts, we architected a visually stunning frontend ecosystem:

 **Command Center Dashboard:** High-level portfolio metrics visually tracking aggregate RED, AMBER, and GREEN distributor volumes.

 **Drag-and-Drop Ingestion:** A seamless UI for analysts to securely upload raw .csv files, complete with real-time FastAPI progress indicators.

 **shadcn/ui Integration:** Built upon Tailwind CSS, the application boasts a premium, glassmorphic aesthetic featuring polished interactive tables and dialogs.

6 Business Deliverables & Output

► 6.1 Automated Risk Cards (.docx)

To bridge the gap between software and operations, we engineered an automated document generation service. It instantly produces branded, beautifully formatted Word-based Risk Cards summarizing a dealer's transaction history, score, and top mitigating Reason Codes.

► 6.2 Executive Pitch Deck

We distilled our technical architecture into a highly refined presentation, translating complex log-odds mathematics into clear ROI and operational efficiency metrics for stakeholders.

7 Detailed Effort & Hours Breakdown

The following table tracks the **28 combined person-hours** invested over the Week 2 sprint. By executing parallel development (Frontend vs. ML Backend), we drastically minimized total hour bloat while maximizing product quality.

Engineering Domain	Key Deliverables & Scope	Hours
Data & Feature Eng.	Synthetic generation, pipeline stress-testing (96% yield), VIF remediation, WOE/IV validation	5.5
ML Model & Edge Cases	5-fold CV Logistic model, Cold-start routing engine, Log-odds explainability	5.0
Backend (FastAPI)	REST schemas, async routers, joblib persistence, pipeline orchestration	4.5
Frontend (Next.js)	React components, shadcn/ui styling, state management, dashboard analytics	8.5
Reporting & Output	DOCX Risk Cards generation, Unified JSON data exports	2.5
Business Strategy	Executive pitch deck, unified changelog synthesis	2.0
Total Cumulative Sprint Effort:		28.0

8 Conclusion

The delivery of the Distributor Credit Risk platform marks a definitive shift from manual intuition to algorithmic precision. By successfully uniting a mathematically robust machine learning core with a state-of-the-art Next.js and FastAPI architecture, we have delivered a comprehensive, enterprise-ready product within a single 28-hour sprint. Highly scalable, visually stunning, and rigorously tested, this platform perfectly positions the firm to execute data-driven credit decisions with absolute confidence.